

EXPERIMENT 6

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Experiment 6 – Implement VLAN and VLAN Trunking Protocols.

Aim: To implement Virtual Local Area Networks (VLANs) and VLAN trunking on network switches, and to configure communication between different VLANs using appropriate trunking protocols.

Objective

- To understand what a VLAN is and why it's used in a network.
- To create more than one VLAN on a switch.
- To assign specific switch ports to specific VLANs.
- To set up a trunk link so multiple VLANs can pass through a single connection between switches.
- To learn how VLAN trunking protocols like IEEE 802.1Q work.
- To check that VLAN separation and communication are working correctly using basic network commands.

Theory:

Layer 2 Switch:

A normal switch (like the Cisco 2960) works using MAC addresses. It keeps a MAC address table so it knows which PC is connected to which port, and forwards data only to the right port. It cannot route between two different networks on its own.

Layer 3 Switch:

A Layer 3 switch can do everything a Layer 2 switch does, plus it can route traffic using IP addresses. This means it can let two different VLANs talk to each other (called Inter-VLAN Routing), something a plain Layer 2 switch cannot do by itself.

VLAN:

VLAN = Virtual LAN. It lets you take one physical switch and split it into multiple separate logical networks. So instead of buying two switches for two departments, you can use one switch and just tell it "these ports belong to Department A" and "these ports belong to Department B."

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Need of VLANs:

Your second screenshot (the one with the crossed wires and a single switch) is a good example of the problem VLANs solve. In that setup, PC1 and PC2 are both meant to be HR, and PC3/PC4 are Sales — but they're all just plugged into one switch with no VLANs configured. That means:

- All four PCs sit in the same broadcast domain.
- Traffic doesn't have any logical separation.
- There's no real security boundary between HR and Sales — anyone could potentially see anyone else's broadcast traffic.

Your first screenshot fixes this: HR (PC1, PC3) and Sales (PC2, PC4) are now split cleanly, connected through two separate switches, and those switches are joined by a **trunk link** so both VLANs can still reach across when needed.

Access Port v/s Trunk Port:

	Access Port	Trunk Port
Carries	One VLAN only	Multiple VLANs
Normally connects	End devices (PCs, printers)	Switch to switch
Example	Fa0/1 → VLAN 10	Gi0/1 carrying VLAN 10 & 20

In your topology (screenshot 1), the link labelled "TRUNK" between the two 2960 switches is exactly this — one cable, carrying both HR's and Sales' traffic at the same time.

How does the other switch know which VLAN a frame belongs to?

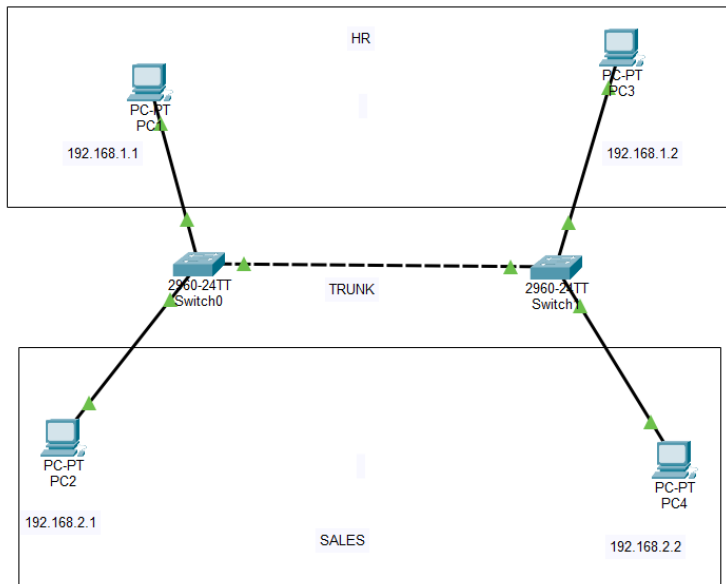
This is where **VLAN tagging** comes in. When a frame crosses the trunk link, the sending switch adds a small tag to it saying "I belong to VLAN 10" (or whichever VLAN). The standard used for this tagging is **IEEE 802.1Q**. The receiving switch reads this tag and forwards the frame only to ports in the correct VLAN.

VLAN Trunking vs VTP :

- **VLAN Trunking** = the general idea of carrying multiple VLANs over one link.
- **802.1Q** = the actual standard used to tag frames on that trunk link.
- **VTP** = a completely different thing — it's used to share VLAN configuration automatically between multiple Cisco switches, not to carry VLAN traffic.

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Practical Steps:



Step 1: Take two switches 2960 Cisco and take 4 PC's.

Step 2: Use Copper-Straight Through cables to connect them.

PC1 → SW1 Fa0/1

PC2 → SW1 Fa0/2

PC3 → SW2 Fa0/1

PC4 → SW2 Fa0/2

Then connect two switches:

SW1 Gi0/1 → SW2 Gi0/1

Step 3:

Assign IP Addresses:

PC1:

IP Address: 192.168.10.11

Subnet Mask: 255.255.255.0

PC3:

IP Address: 192.168.10.12

Subnet Mask: 255.255.255.0

PC1 and PC3 are going to belong to: VLAN 10

So their IP addresses are also from: 192.168.10.0

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PC2:

IP Address: 192.168.20.11

Subnet Mask: 255.255.255.0

PC4:

IP Address: 192.168.20.12

Subnet Mask: 255.255.255.0

VLAN 20.

Step 4: Configuring VLAN on SW1 (Switch 1).

SW1 -> GO TO CLI.

Switch>

Enter Privileged Mode:

Type: Enable

Switch#

Enter Configuration Mode:

Type: Configure Terminal or Conf t

Switch(config)#

Step 5: Create VLAN 10 on SW1

Type: VLAN 10

Switch(config-vlan)#

Give it a name: name STUDENTS

Now VLAN 10 has been created.

VLAN 10 -> STUDENTS.

Step 6: Create VLAN 20 on SW1

Type: Exit

Then: VLAN 20

And Name FACULTY

Now SW1 has:

VLAN 10 -> STUDENTS

VLAN 20 -> FACULTY

Step 7: Assigning PORTS to VLAN.

Creating a VLAN does not automatically put PCs into that VLAN.

We still have to tell the switch: "Fa0/1 belongs to VLAN 10"

And "Fa0/1 belongs to VLAN 20"

Step 8: Configure Fa0/1 as an Access Port.

Type: Exit

Now enter the interface: interface fa0/1

You should see: Switch(config-if)#

Now tell the switch that this is an access port:

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Type: Switchport mode access

Then assign it to VLAN 10: switchport access vlan 10

So the complete configuration will be:

```
interface fa0/1
```

```
switchport mode access
```

```
switchport access vlan 10
```

By this we tell SW1 that fa0/1 -> Access Port -> VLAN 10.

Therefore, PC1 -> Fa0/1 -> VLAN 10.

Step 9: Configure Fa0/2 for VLAN 20:

Type: Exit

```
interface fa0/2
```

```
then switchport mode access
```

```
then switchport access vlan 10.
```

Step 10: Repeat the same process for SW2.

Step 11: Now we have to configure the TRUNK.

This is the most important part of the experiment.

We have: SW1 Gi0/1 ===== Gi0/1 SW2

Currently, this is just a normal switch-to-switch connection.

We need to tell both switches:

"This link is going to carry multiple VLANs."

Therefore, we configure it as a trunk.

Step 12: On SW1:

```
enable
```

```
configure terminal
```

```
then interface gigabitEthernet 0/1
```

```
then switchport mode trunk
```

Now Gig0/1 is now the trunk port.

Step 13: Repeat the same process for SW2.

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Learning Outcome

After this experiment, you should be able to:

1. Explain the difference between a Layer 2 and Layer 3 switch.
2. Create and assign VLANs on a switch.
3. Set up a trunk link to carry multiple VLANs between two switches.
4. Understand why VLAN tagging (802.1Q) is needed on trunk links.